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Markov-Switching Models

Gabriel Rodríguez
Pontificia Universidad Católica del Perú

Introduction

- A class of nonlinear models where there are more than one regime.
- The determination of the regimes depends of an unobserved variable.
- Basic references: Hamilton (1989), Kim and Nelson (1999) for univariate Markov-Switching models and Krolzig (1997) for Multivariate Markov-Switching Models.

Brief Motivation

- A possible model (see graph):

$$
\begin{aligned}
y_t - \mu_1 &= \phi(y_{t-1} - \mu_1) + \epsilon_t, \qquad \text{before 1982} \\
y_t - \mu_2 &= \phi(y_{t-1} - \mu_2) + \epsilon_t, \qquad \text{after 1982}
\end{aligned}
$$

  with $\mu_2 < \mu_1$. Issue: incomplete specification. The change in regime is itself a random variable.

- A complete specification would therefore include a description of the probability law governing the change from μ_1 to μ_2. Let s_t be an unobserved variable named *state* or *regime* that the process was in at date t.
- Last two equations may be written as

$$y_t - \mu_{s_t} = \phi(y_{t-1} - \mu_{s_{t-1}}) + \epsilon_t,$$

  for $t = 1, 2, ..., T$, and where $\mu_{s_t} = \mu_1$ if $s_t = 1$; and $\mu_{s_t} = \mu_2$ if $s_t = 2$.

Maximum Likelihood (ML) in Linear Models

- Assume a Gaussian AR(1) process

$$y_t = v + \phi y_{t-1} + \epsilon_t$$

  with $\epsilon_t \sim N(0, \sigma^2)$. Here: $\theta = (v, \phi, \sigma^2)'$.

- We know that $E(y_1) = \mu = v/(1-\phi)$, and $var(y_1) = \sigma^2/(1-\phi^2)$. Therefore:

$$f(y_1; \theta) = \frac{1}{\sqrt{2\pi}\sigma_{y_1}} \exp\{-\frac{(y_1 - \mu_{y_1})^2}{2\sigma_{y_1}^2}\}$$

- On another hand,

$$\begin{aligned} f(y_2|y_1; \theta) &= \frac{1}{\sqrt{2\pi}\sigma} \exp\{-\frac{(y_2 - v - \phi y_1)^2}{2\sigma^2} \\ f(y_3|y_2, y_1; \theta) &= \frac{1}{\sqrt{2\pi}\sigma} \exp\{-\frac{(y_3 - v - \phi y_2)^2}{2\sigma^2} \end{aligned}$$

- Consequently:

$$
\begin{aligned}
f(y_t|y_{t-1}, y_{t-2}, ..., y_1; \theta) &= f(y_t|y_{t-1}; \theta) \\
&= \frac{1}{\sqrt{2\pi}\sigma} \exp\{-\frac{(y_t - v - \phi y_{t-1})^2}{2\sigma^2}
\end{aligned}
$$

- Thus, the joint density of the first t observations is

$$f(y_t, y_{t-1}, ..., y_1; \theta) = f(y_t|y_{t-1}; \theta) \times f(y_{t-1}, y_{t-2}, ..., y_1; \theta)$$

- The likelihood of the complete sample is calculate by

$$f(y_T, y_{T-1}, ..., y_1; \theta) = \prod_{t=2}^{T} f(y_t|y_{t-1}; \theta) \times f(y_1; \theta).$$

- Then, the log likelihood function is:

$$L(\theta) = \log f(y_1; \theta) + \sum_{t=2}^{T} \log(y_t|y_{t-1}; \theta).$$

- Maximize $L(\theta)$ to obtain $\widehat{\theta}$.

Markov Chains

- Assume the discrete variable s_t. Let $s_t \in \{1, 2, ..., M\}$. Then,

$$P\{s_t = j|s_{t-1} = i, s_{t-2} = k, ...\} = \Pr[s_t = j|s_{t-1} = i] = p_{ij}$$

- Last expression is an M-state Markov chain with transition probabilities $\{p_{ij}\}$, $\forall\ i, j = 1, 2, .., M$, and with condition $\sum_{j=1}^{M} p_{ij} = 1$. The transition matrix is

$$P = \begin{bmatrix} p_{11} & p_{12} & ... & p_{1M} \\ p_{21} & p_{22} & ... & p_{2M} \\ .... & .... & ... & .... \\ p_{M1} & p_{M2} & ... & P_{MM} \end{bmatrix}.$$

Statistical Analysis of $i.i.d.$ Mixture Distributions

- $i.i.d.$ mixture distributions is a particular case of the MS models where the regime itself is described as the outcome of an unobserved Markov chain.

- Recall s_t defined before. When $s_t = 1$, y_t is assumed to have been drawn from a $N(\mu_1, \sigma_1^2)$. When $s_t = 2$, y_t is assumed to have been drawn from a $N(\mu_2, \sigma_2^2)$. Therefore, the density of y_t conditional on $s_t = j$ is
$$f(y_t|s_t = j; \theta) = \frac{1}{\sqrt{2\pi}\sigma_j} \exp\{-\frac{(y_t - \mu_j)^2}{2\sigma_j^2}\}$$
for $j = 1, 2, ..., M$, and $\theta = (\mu_1, ..., \mu_M; \sigma_1^2, ..., \sigma_M^2)'$.

- The unobserved regime $\{s_t\}$ is presumed to have been generated by some probability distribution, for which the unconditional probability that $s_t = j$ is π: $\Pr\{s_t = j; \theta\} = \pi_j$, for $j = 1, 2, ..., M$. Note that θ also includes $\pi_1, ..., \pi_M$.

- Recall that $\Pr(A|B) = \Pr(A \text{ and } B)/P(B)$, provided that $P(B) \neq 0$. Then $\Pr(A \text{ and } B) = P(A|B) \times P(B)$.

- Therefore:
$$p(y_t, s_t = j; \theta) = f(y_t|s_t = j; \theta) \times \Pr(s_t = j; \theta)$$

- Last expression is the joint density-distribution function of y_t and s_t. This expresion may be written by
$$p(y_t, s_t = j; \theta) = \frac{\pi_j}{\sqrt{2\pi}\sigma_j} \exp\{-\frac{(y_t - \mu_j)^2}{2\sigma_j^2}\}.$$

- The unconditional density of y_t can be found by summing last expression over all possible values of j:

$$\begin{aligned} f(y_t;\theta) &= \sum_{j=1}^{M} p(y_t, s_t = j;\theta) \\ &= \frac{\pi_1}{\sqrt{2\pi}\sigma_1}\exp\{-\frac{(y_t-\mu_1)^2}{2\sigma_1^2}\} \\ &\quad +\ldots\ldots \\ &\quad +\frac{\pi_M}{\sqrt{2\pi}\sigma_M}\exp\{-\frac{(y_t-\mu_M)^2}{2\sigma_M^2}\}. \end{aligned}$$

- Last expression is the relevant density distribution of y_t. If s_t is $i.i.d.$ across different dates t, then the log likelihood for the observed data may be calculated by

$$L(\theta) = \sum_{t=1}^{T} \log f(y_t;\theta).$$

Univariate Markov Switching Models

- Assume a model with structural breaks in the parameters:

$$\begin{aligned} y_t &= x_t\beta_{s_t} + e_t \\ e_t &\sim N(0, \sigma^2_{s_t}), \\ \beta_{s_t} &= \beta_0(1 - s_t) + \beta_1 s_t, \\ \sigma^2_{s_t} &= \sigma^2_0(1 - s_t) + \sigma^2_1 s_t, \end{aligned}$$

  $t = 1, 2, ..., T$, and $s_t = 0$ or 1. Therefore, under regime 1, parameters are given by β_1 and σ^2_1, and under regime 0, parameters are given by β_0 and σ^2_0.

- A major problem with the above model arises when s_t is not observed. In this case (rule of thumb), we use the following two steps to determine the log likelihood function:

  - Step 1: Consider the joint density of y_t and the unobserved s_t variable which is the product of the conditional and marginal densities:

  $$f(y_t, s_t|\Omega_{t-1}) = f(y_t|s_t, \Omega_{t-1}) \times f(s_t|\Omega_{t-1})$$

    where Ω_{t-1} is the set of information up to time $t - 1$.

  - Step 2: To obtain the marginal density of y_t, integrate the s_t variable out of the above joint density by summing over all possible values of s_t:

$$\begin{aligned}
f(y_t|\Omega_{t-1}) &= \sum_{s_t=0}^{1} f(y_t, s_t|\Omega_{t-1}) \\
&= \sum_{s_t=0}^{1} f(y_t|s_t, \Omega_{t-1}) \times f(s_t|\Omega_{t-1}) \\
&= \frac{1}{2\pi\sigma_0^2} \exp(-\frac{[y_t - x_t\beta_0]^2}{2\sigma_0^2} \times \Pr[s_t = 0|\Omega_{t-1}] \\
+\frac{1}{2\pi\sigma_1^2} \exp(-\frac{[y_t - x_t\beta_1]^2}{2\sigma_1^2} \times \Pr[s_t &= 1|\Omega_{t-1}]
\end{aligned}$$

The log likelihood function is given by

$$\ln L = \sum_{t=1}^{T} \ln\{\sum_{s_t=0}^{1} f(y_t|s_t, \Omega_{t-1}) \times f(s_t|\Omega_{t-1})\}.$$

- The marginal density given above may be interpreted as a weighted average of the conditional densities given $s_t = 0$ and $s_t = 1$, respectively.
- To calculate the marginal density of y_t and thus, the log likelihood function, we need to calculate the weighting factors: $\Pr[s_t = 0|\Omega_{t-1}]$ and $\Pr[s_t = 1|\Omega_{t-1}]$. Without a priori assumptions about the stochastic behavior of the s_t variable, this will not be possible.
- The evolution of the discrete variable s_t may be dependent of $s_{t-1}, s_{t-2}, ..., s_{t-r}$. In this case the process of s_t is named as an r-th order Markov-switching process. Consider the case of a two-state, first order Markov switching process for s_t with the following transition probabilities:

$$\begin{aligned}
\Pr[s_t &= 1|s_{t-1} = 1] = p = \frac{\exp(p_0)}{1 + \exp(p_0)}, \\
\Pr[s_t &= 0|s_{t-1} = 0] = q = \frac{\exp(q_0)}{1 + \exp(q_0)}.
\end{aligned}$$

- Calculation of the weighting terms $Pr[s_t = j|\Omega_{t-1}]$, $j = 0, 1$ is not as straightforward as before. We adopt the following filter for the calculation of the weighting terms:

  - Step 1: Given $Pr[s_{t-1} = i|\Omega_{t-1}]$, $i = 0, 1$, at the beginning of time t or the t-th iteration, the weighting terms $Pr[s_t = j|\Omega_{t-1}]$, $j = 0, 1$ are calculated as

$$\begin{aligned}
\Pr[s_t &= j|\Omega_{t-1}] = \sum_{i=0}^{1} \Pr[s_t = j, s_{t-1} = i|\Omega_{t-1}] \\
&= \sum_{i=0}^{1} \Pr[s_t = j|s_{t-1} = i] \times \Pr[s_{t-1} = i|\Omega_{t-1}],
\end{aligned}$$

    where $Pr[s_t = j|s_{t-1} = i]$, $i = 0, 1$, $j = 0, 1$, are the transition probabilities.

  - Step 2: Once y_t is observed at the end of time t, or at the end of the t-th iteration, we can update the probability term in the following way:

$$\begin{aligned}
\Pr[s_t &= j|\Omega_t] = \Pr[s_t = j|\Omega_{t-1}, y_t] \\
&= \frac{f(s_t = j, y_t|\Omega_{t-1})}{f(y_t|\Omega_{t-1})} \\
&= \frac{f(y_t|s_t = j, \Omega_{t-1}) \times \Pr[s_t = j|\Omega_{t-1}]}{\sum_{i=0}^{1} f(y_t|s_t = j, \Omega_{t-1}) \times \Pr[s_t = j|\Omega_{t-1}]}
\end{aligned}$$

    where $\Omega_t = \{\Omega_{t-1}, y_t\}$.

- The above steps may be iterated to get $Pr[s_t = j|\Omega_{t-1}]$, $t = 1, 2, ..., T$. To start the filter at time $t = 1$, we need $\Pr[s_0|\Omega_0]$. We may employ the following steady-state or unconditional probabilities of s_t:

$$\begin{aligned}
\pi_0 &= \Pr[s_0 = 0|\Omega_0] = \frac{1-p}{2-p-q} \\
\pi_1 &= \Pr[s_0 = 1|\Omega_0] = \frac{1-q}{2-p-q}.
\end{aligned}$$

- In general, an autoregressive model of order k with first-order, M-state Markov-switching mean and variance may be written as

$$\begin{aligned}
\phi(L)(y_t - \mu_{s_t}) &= e_t, \\
& \quad e_t \sim N(0, \sigma^2_{s_t}), \\
\Pr[s_t &= j | s_{t-1} = i] = p_{ij}, \qquad i, j = 1, 2, ..., M, \\
\sum_{j=1}^{M} p_{ij} &= 1, \\
\mu_{s_t} &= \mu_1 s_{1t} + \mu_2 s_{2t} + ... + \mu_M s_{Mt}, \\
\sigma^2_{s_t} &= \sigma^2_1 s_{1t} + \sigma^2_2 s_{2t} + ... + \sigma^2_M s_{Mt},
\end{aligned}$$

where $s_{mt} = 1$, if $s_t = m$, and $s_{mt} = 0$, otherwise.

- Consider the simple case where we have an $AR(1)$ model:

$$\begin{aligned}
(y_t - \mu_{s_t}) &= \phi_1(y_{t-1} - \mu_{s_{t-1}}) + e_t, \\
& \quad e_t \sim i.i.d. \quad N(0, \sigma^2_{s_t}).
\end{aligned}$$

- Now, in writing the density of y_t given past information Ω_{t-1}, we need s_t and s_{t-1}, which are not observed. We proceed in the same way as before, except instead of considering a joint density of y_t and s_t, we now need consider the joint density of y_t, s_t and s_{t-1}:

  – Step 1: Derive the joint density of y_t, s_t and s_{t-1}, conditional on past information Ω_{t-1}:

$$f(y_t, s_t, s_{t-1} | \Omega_{t-1}) = f(y_t | s_t, s_{t-1}, \Omega_{t-1}) \times \Pr[s_t, s_{t-1} | \Omega_{t-1}],$$

  where

$$f(y_t | s_t, s_{t-1}, \Omega_{t-1}) = \frac{1}{\sqrt{2\pi\sigma^2_{s_t}}} \exp(-\frac{\{(y_t - \mu_{s_t}) - \phi_1(y_{t-1} - \mu_{s_{t-1}})\}^2}{2\sigma^2_{s_t}}.$$

– Step 2: To get $f(y_t|\Omega_{t-1})$, integrate s_t and s_{t-1} out of the joint density by summing the joint density over all possible values of s_t and s_{t-1}:

$$\begin{aligned} f(y_t|\Omega_{t-1}) &= \sum_{s_t=1}^{M}\sum_{s_{t-1}=1}^{M} f(y_t, s_t, s_{t-1}|\Omega_{t-1}) \\ &= \sum_{s_t=1}^{M}\sum_{s_{t-1}=1}^{M} f(y_t|s_t, s_{t-1}, \Omega_{t-1}) \times \Pr[s_t, s_{t-1}|\Omega_{t-1}], \end{aligned}$$

where the marginal density $f(y_t|\Omega_{t-1})$ is a weighted average of M^2 conditional densities, weights being $Pr[s_t = j, s_{t-1} = i|\Omega_{t-1}]$, $i = 1, 2, ..., M$, and $j = 1, 2, ..., M$.

- Then the log likelihood function is given by:

$$\ln L = \sum_{t=1}^{T} \ln\{\sum_{s_t=1}^{M}\sum_{s_{t-1}=1}^{M} f(y_t|s_t, s_{t-1}, \Omega_{t-1}) \times \Pr[s_t, s_{t-1}|\Omega_{t-1}]\}.$$

- To complete the above procedure, we still need to deal with the problem of calculating $Pr[s_t = j, s_{t-1} = i|\Omega_{t-1}]$. The procedure to obtain this is named *Filtering*. It is done in two steps.

  – Step 1: Given $Pr[s_{t-1} = i|\Omega_{t-1}]$, $i = 1, 2, ..., M$, at the beginning of time t or the t-th iteration, the weighting terms $Pr[s_t = j, s_{t-1} = i|\Omega_{t-1}]$, $i = 1, 2, ..., M$, $j = 1, 2, ..., M$, are calculated as

$$\Pr[s_t = j, s_{t-1} = i|\Omega_{t-1}] = \Pr[s_t = j|s_{t-1} = i] \times \Pr[s_{t-1} = i|\Omega_{t-1}],$$

  where $Pr[s_t = j|s_{t-1} = i]$, $i = 1, 2, ..., M$, and $j = 1, 2, ..., M$, are the transition probabilities.

  – Step 2: Once y_t is observed at the end of time t, or at the end of the t-th iteration, we can update the probability terms in the following way:

$$
\begin{aligned}
\Pr[s_t &= j, s_{t-1} = i|\Omega_t] = \Pr[s_t = j, s_{t-1} = i|\Omega_{t-1}, y_t] \\
&= \frac{f(s_t = j, s_{t-1} = i, y_t|\Omega_{t-1})}{f(y_t|\Omega_{t-1})} \\
&= \frac{f(y_t|s_t = j, s_{t-1} = i, \Omega_{t-1}) \times \Pr[s_t = j, s_{t-1} = i|\Omega_{t-1}]}{\sum_{s_t=1}^{M} \sum_{s_{t-1}=1}^{M} f(y_t|s_t = j, s_{t-1} = i, \Omega_{t-1}) \times \Pr[s_t = j, s_{t-1} = i|\Omega_{t-1}]},
\end{aligned}
$$

with

$$\Pr[s_t = j|\Omega_t] = \sum_{s_{t-1}=1}^{M} \Pr[s_t = j, s_{t-1} = i|\Omega_t].$$

- Iterating the above two steps for $t = 1, 2, ..., T$ provides us with the appropriate weighting terms. To start the above filter at time $t = 1$, we may employ the steady-state or unconditional probabilities. For a two-state, first-order Markov switching, the steady-state probabilities are given by

$$
\begin{aligned}
\pi_1 &= \Pr[s_0 = 1|\Omega_0] = \frac{1 - p_{22}}{2 - p_{22} - p_{11}}, \\
\pi_2 &= \Pr[s_0 = 2|\Omega_0] = \frac{1 - p_{11}}{2 - p_{22} - p_{11}}.
\end{aligned}
$$

Expected Duration of a Regime in a Markov-Switching Model

- The diagonal elements of the matrix of the transition probabilities contain information on the expected duration of a state of regime. Question: Given that we are currently in regime j or state j ($s_t = j$), how long, on average, will the regime j last?

- We define D as the duration of state j:

  $D = 1$ if $s_t = j$ and $s_{t+1} \neq j$; $Pr[D = 1] = (1 - p_{jj})$

  $D = 2$, if $s_t = s_{t+1} = j$ and $s_{t+2} \neq j$; $Pr[D = 2] = p_{jj}(1 - p_{jj})$

  $D = 3$, if $s_t = s_{t+1} = s_{t+2} = j$ and $s_{t+3} \neq j$; $Pr[D = 3] = p_{jj}^2(1 - p_{jj})$

  $D = 4$, if $s_t = s_{t+1} = s_{t+2} = s_{t+3} = j$ and $s_{t+4} \neq j$; $Pr[D = 4] = p_{jj}^3(1 - p_{jj})$

  and so on. Then, the expected duration of regime j may be derived as

$$
\begin{aligned}
E(D) &= \sum_{j=1}^{\infty} j \Pr[D = j] \\
&= 1 \times \Pr[s_{t+1} \neq j | s_t = j] \\
+2 \times \Pr[s_{t+1} &= j, s_{t+2} \neq j | s_t = j] \\
+3 \times \Pr[s_{t+1} &= j, s_{t+2} = j, s_{t+3} \neq j | s_t = j] \\
+4 \times \Pr[s_{t+1} &= j, s_{t+2} = j, s_{t+3} = j, s_{t+4} \neq j | s_t = j] \\
& \quad +.... \\
&= 1 \times (1 - p_{jj}) + 2 \times p_{jj}(1 - p_{jj}) + 3 \times p_{jj}^2(1 - p_{jj}) + ... \\
&= \frac{1}{1 - p_{jj}}.
\end{aligned}
$$

Example 1. Hamilton's Markov-Switching Model of Business Fluctuations

- In Markov-switching models of business fluctuations, the turning point is treated as a structural event that is inherent in the data-generating process.

- These models capture a particular form of nonlinear dynamics or asymmetry in the business fluctuations. Hamilton (1989) allows the mean of the growth in real GNP to be evolving according to a two-state Markov-switching process. Therefore, the dynamics of recessions are qualitatively distinct from those of expansions.

- The growth in real GNP is modeled as an AR(4) process:

$$
\begin{aligned}
(\Delta y_t - \mu_{s_t}) &= \phi_1(\Delta y_{t-1} - \mu_{s_{t-1}}) \\
&\quad + \phi_2(\Delta y_{t-2} - \mu_{s_{t-2}}) + ... + \phi_4(\Delta y_{t-4} - \mu_{s_{t-4}}) + e_t, \\
&\quad e_t \sim i.i.d. \quad N(0, \sigma^2), \\
\mu_{s_t} &= \mu_0(1 - s_t) + \mu_1 s_t, \\
\Pr[s_t &= 1|s_{t-1} = 1] = p, \\
\Pr[s_t &= 0|s_{t-1} = 0] = q,
\end{aligned}
$$

  where roots of $\phi(L) = (1 - \phi_1 L - ... - \phi_4 L^4) = 0$ lie outside the unit circle and y_t is the log of real GDP or real GNP.

- Sample 1952:2-1984:4

- Filtered and smoothed probabilities are in close agreement with the NBER reference cycle.

- When the sample is extended to 1952:2-1995:3, the model fails to provide reasonable inferences on the probabilities of a recession or a boom. Potential reasons: (i) lack of mechanism to account for a productivity slowdown in the more recent sample. In other words, the model assumes that the average growth rate of output during a boom or recession is the same over the entire sample; (ii) even in the absence of a productivity slowdown, the monetary policy may have had more stabilizing effect on the economic activity in recent years. Therefore, the gap between average growth rates of output during boom and recession would be smaller than in the earlier sample.

- To account for the above possibilities, the following equation is used:

$$\mu_{s_t} = (\mu_0 + \mu_0^* D_t)(1 - s_t) + (\mu_1 + \mu_1^* D_t)s_t,$$

  where D_t is a dummy variable set equal to 1 for the subsample 1983:1-1995:3, and 0 for an earlier sample period.

- The inclusion of a dummy variable potentially captures a change in the mean growth rates during boom or recession.

- Now, the probabilities are in close agreement with the NBER reference cycle.

Example 2. A Three-State Markov-Switching Model of the Real Interest Rate

- The work of Fama (1975) about the efficiency of the bill market rests on the postulate that the ex ante real interest rate is constant.

- Garbade and Wachtel (1978), Nelson and Schwert (1977) base their analysis on the assumption that the real rate follows an integrated process.

- A finding that the ex ante real interest rate has a unit root would have important implications concerning tha fact that the real rate be used as a guide to the conduct of monetary policy (Walsh, 1987).

- Literature on the tests of the stationarity of the real interest rate gives mixed results. Rose (1988) and Walsh (1987) cannot reject the null hypothesis of a unit root. However, Perron (1990) rejects the unit root hypothesis after introduce regime shifts. He argues that the unit root statistics are biased toward nonrejection of the unit root hypothesis when the series contains a sudden change in the mean.

- Garcia and Perron (1996) use the model of Hamilton (1989) to explicitly account for regime shifts in an autoregressive model of the ex post real interest rate. They employ the following AR(2) process with a three-state Markov-switching in mean and variance:

$$\begin{aligned}
(y_t - \mu_{s_t}) &= \phi_1(y_{t-1} - \mu_{s_{t-1}}) + \phi_2(y_{t-2} - \mu_{s_{t-2}}) + e_t, \\
e_t &\sim N(0, \sigma^2 s_t), \\
\mu_{s_t} &= \mu_1 s_{1t} + \mu_2 s_{2t} + \mu_3 s_{3t}, \\
\sigma^2_{s_t} &= \sigma^2_1 s_{1t} + \sigma^2_2 s_{2t} + \sigma^2_3 s_{3t}, \\
p_{ij} &= \Pr[s_t = j | s_{t-1} = i], \\
\sum_{j=1}^{3} p_{ij} &= 1,
\end{aligned}$$

  where $s_{jt} = 1$, if $s_t = j$, and $s_{jt} = 0$, otherwise, $j = 1, 2, 3$, and where y_t is the ex post real interest rate calculated by substracting the (CPI) inflation rate from the nominal interest rate (three-month Treasury bill rate).

- Sample: quarterly data for 1960:1-1990:4.

- Conjecture of Perron (1990) is that if the model is given by the above model but a linear model is estimated, then the sum of the autoregressive coefficients will be biased toward unity even when the true sum is close to zero.

- A linear $AR(4)$ model is estimated. Sum of the autoregressive coefficients is 0.841.

- In the Markov-switching model the sum of the autoregressive coefficients is close to zero.

- The ex ante real interest rate for the $AR(2)$ proces is estimated based on the following expression

$$E(y_t|\widetilde{y}_{t-1}) = E(\mu_{s_t}|\widetilde{y}_t) + \phi_1 E(y_{t-1} - \mu_{s_{t-1}}|\widetilde{y}_t) + \phi_2 E(y_{t-2} - \mu_{s_{t-2}}|\widetilde{y}_t),$$

  where $\widetilde{y}_t = [y_1, ..., y_t]$. Under the assumption of rational expectations, the above expression may be interpreted as the ex ante real interest rate. It is because the only difference between this expression and the ex post interest rate is a white noise "forecast" error.

- Garcia and Perron (1996) use various tests that suggest that the real interest rate may not have a unit root. Instead, shocks to the real interest rate may be temporary in nature with tendency to revert to some mean value, which is subject to infrequent shifts.

Specification of Markov-Switching-VAR Models (MS-VAR)

- Basic reference: Krolzig (1997).
- Consider the following linear $VAR(p)$ model:

$$y_t = v + A_1 y_{t-1} + ... + A_p y_{t-p} + u_t \tag{1}$$

where $u_t \sim IID(0, \Sigma)$, $y_0, y_1, ..., y_{-p}$ are fixed, $A(L) = I_k - A_1 L - ... - A_p L^p$ is a $(k \times k)$ dimensional lag polynomial with $|A(z)| \neq 0$ for $|z| \leq 1$.

- If $u_t \sim NID(0, \Sigma)$, the model (1) is the intercept form of a Gaussian $VAR(p)$. Model (1) may be written in the form of a mean-adjusted $VAR(p)$:

$$y_t - \mu = A_1(y_{t-1} - \mu) + ... + A_p(y_{t-p} - \mu) + u_t \tag{2}$$

where $\mu = (I_k - \sum_{j=1}^{p} A_j)^{-1} v$, is a $(k \times 1)$ dimensional mean of y_t.

- The process is completed by specifying the generating process for s_t: $p_{ij} = Pr[s_t = j | s_{t-1} = i]$, $\sum_{j=1}^{M} p_{ij} = 1$, $\forall\ i, j \in \{1, 2, ..., M\}$. In other words:

$$P = \begin{bmatrix} p_{11} & p_{12} & ... & p_{1M} \\ p_{21} & p_{22} & ... & p_{2M} \\ .... & .... & ... & ... \\ p_{M1} & p_{M2} & ... & P_{MM} \end{bmatrix} \tag{3}$$

with $p_{iM} = 1 - p_{i1} - p_{i2} - ... - p_{iM-1}$, for $i = 1, 2, ..., M$.

- Then, using (2) and the specification of the generating process for s_t, we have a Markov-Switching $VAR(p)$ with M regimes:

$$y_t - \mu(s_t) = A_1(s_t)(y_{t-1} - \mu(s_t)) + ... + A_p(s_t)(y_{t-p} - \mu(s_{t-p})) + u_t. \quad (4)$$

where $u_t \sim NID(0, \Sigma(s_t))$. This model is useful in cases where one-time jump is observed in the time series. When the mean changes smoothly to a new level, the preferred model is

$$y_t = v(s_t) + A_1(s_t)y_{t-1} + ... + A_p(s_t)y_{t-p} + u_t \quad (5)$$

- Models specified in (4) and (5) are not equivalent. They imply different dynamics of adjustemnt of the variables after a change in regime.

- Summary of MS-VAR Models

		MSM		MSI	
		μ varying	μ invariant	v varying	v invariant
A_j invariant	Σ invariant	MSM-VAR	linear MVAR	MSI-VAR	linear VAR
	Σ varying	MSMH-VAR	MSH-VAR	MSIH-VAR	MSH-VAR
A_j variant	Σ invariant	MSMA-VAR	MSA-VAR	MSIA-VAR	MSA-VAR
	Σ varying	MSMAH-VAR	MSAH-VAR	MSIAH-VAR	MSAH-VAR

- Notation

  - M: Markov-Switching *mean*,
  - I: Markov-Switching *intercept*,
  - A: Markov-Switching *autoregressive parameters*,
  - H: Markov-Switching *heteroskedasticity*.

Likelihood-based Statistical Issues

- Until now, we have i) the Gaussian VAR model as the conditional data generating process; ii) the Markov chain as the regime generating process. We present a brief sketch of the likelihood-based statistical methods.

- For a given regime s_t and lagged endogenous variables: $Y_{t-1} = (y_{t-1}, y_{t-2}, ..., y_1, y_0, ..., y_{1-p})$, the conditional probability density function is denoted by $p(y_t|s_t, Y_{t-1})$. It is also assumed (in (4) and (5)) that u_t is normally distributed.

- Define ξ_t as a $(M \times 1)$ vector whose jth element is equal to unity if $s_t = j$ and whose jth element equals zero otherwise. Thus, for example, when $s_t = 1$, ξ_t is equal to the first column of the identity matrix I_M. When $s_t = 2$, the vector ξ_t is the second column of I_M; and so on:

$$\xi_t = \left\{ \begin{array}{cc} (1,0,0,...,0)' & \text{when } s_t = 1 \\ (0,1,0,...,0)' & \text{when } s_t = 2 \\ & \vdots \\ (0,0,0,...,1)' & \text{when } s_t = M \end{array} \right\}. \tag{6}$$

- If $s_t = i$, then the jth element of ξ_{t+1} is a random variable that takes on the value unity with probability p_{ij} and takes on the value zero otherwise. Such a random variable has expectation p_{ij}. Then

$$E(\xi_{t+1}|s_t = i) = \begin{bmatrix} p_{i1} \\ p_{i2} \\ \vdots \\ p_{iM} \end{bmatrix}. \tag{7}$$

- The vector in (7) is simply the ith column of the matrix P in (3). Furthermore, when $s_t = i$, the vector ξ_t corresponds to the ith column of I_M, in which case the vector in (7) could be described as $P\xi_t$. Hence:

$$E(\xi_{t+1}|\xi_t) = P\xi_t \tag{8}$$

and by the Markov property, it follows that

$$E(\xi_{t+1}|\xi_t, \xi_{t-1}, ...) = P\xi_t.$$

- In a similar way, the densities of y_t conditional on s_t and Y_{t-1} can be collected to the vector η_t :

$$\eta_t = \begin{bmatrix} p(y_t|\xi_t = 1; Y_{t-1}) \\ \vdots \\ p(y_t|\xi_t = M; Y_{t-1}) \end{bmatrix}. \tag{9}$$

- Consequently,

$$p(y_t|\xi_{t-1}, Y_{t-1}) = \eta_t' P' \xi_{t-1}. \tag{10}$$

- Problem: the regime is assumed unobserved and then, the unobserved regime vector ξ_t has to be replaced by the inference $\Pr(\xi_t|Y_t)$. These probabilities of being in regime m given an information set Y_t, are denoted $\xi_{mt|t}$ and collected in the vector $\widehat{\xi}_{t|t}$:

$$\widehat{\xi}_{t|t} = \begin{bmatrix} p(s_t = 1|Y_t) \\ \vdots \\ p(s_t = M|Y_t) \end{bmatrix}, \tag{11}$$

where, as a consequance of the binarity of the elements of ξ_t, it implies $E(\xi_{mt}) = \Pr(\xi_{mt} = 1) = \Pr(s_t = m)$.

- Thus, the conditional probability density of y_t based upon Y_{t-1} is given by:

$$
\begin{aligned}
p(y_t|Y_{t-1}) &= \int p(y_t, \xi_{t-1}|Y_{t-1}) d\xi_{t-1} \\
&= \int p(y_t|\xi_{t-1}, Y_{t-1}) \times \Pr(\xi_{t-1}|Y_{t-1}) d\xi_{t-1} \\
&= \eta_t' P' \widehat{\xi}_{t-1|t-1} \qquad (12)
\end{aligned}
$$

- Sketch of the technique of setting-up the likelihood function. For given presample values Y_0, the density of the sample $Y = Y_T$ conditional on the states ξ is determined by

$$p(Y|\xi) = \prod_{t=1}^{T} p(y_t|\xi_t, Y_{t-1}). \qquad (13)$$

- The joint probability distribution of observations and states may be calculated as

$$p(Y, \xi) = p(Y|\xi) \times \Pr(\xi) = \prod_{t=1}^{T} p(y_t|\xi_t, Y_{t-1}) \prod_{t=2}^{T} \Pr(\xi_t|\xi_{t-1}) \times \Pr(\xi_1). \qquad (14)$$

- Thus, the unconditional density of Y is given by the marginal density

$$p(Y) = \int p(Y, \xi) d\xi. \qquad (15)$$

The EM Algorithm

- The maximization of the likelihood function of an MS-VAR model entails an iterative technique. This technique gives us the parameters of the autoregression and the transition probabilities governing the Markov chain of the unobserved states. Denote this parameter vector by $\lambda = (\theta, \rho)$.

- ML estimation of the model is based on the Expectation Maximization (EM) algorithm proposed by Hamilton (1990). Each iteration of the EM algorithm consists of two steps:

  - The *expectation* step. It involves a pass through the filtering and smothing algorithms, using the estimated parameter vector $\lambda^{(j-1)}$ of the last *maximization* step in place of the unknown true parameter vector. This delivers an estimate of the smoothed probabilities $\Pr(\xi|Y, \lambda^{(j-1)})$ of the unobserved states ξ_t (which records the history of the Markov chain).

  - The *maximization* step. In this step, an estimate of λ is derived as a solution of the first-order conditions associated with the likelihood function, where the conditional probabilities $\Pr(\xi|Y, \lambda)$ are replaced with the smoothed probabilities $\Pr(\xi|Y, \lambda^{(j-1)})$ derived in last *expectation* step. With the new paremeter vector λ the filter and smoothed probabilities are updated in the next *expectation* step, and so on.

- Regimes constructed in this way constitute an optimal inference on the latent state of the economic process. It follows by the definition of the conditional density that the conditional distribution of the total regime vector ξ is given by

$$\Pr(\xi|Y) = \frac{p(Y,\xi)}{p(Y)}. \tag{16}$$

- The required conditional regime probabilities $\Pr(\xi_t|Y)$ can be derived by marginalization of $\Pr(\xi|Y)$. Calculations are simplified using the recursive and smoothing algorithms discussed in Krolzig (1997). With that, inference for ξ_t given a specified observation set Y_τ, $\tau \leq T$ is possible. It reconstructs the time path of the regime $\{\xi_t\}_{t=1}^{T}$, under alternative information sets:

  - *predicted* regime probabilities $\widehat{\xi}_{t|\tau}$ when $\tau < t$
  - *filtered* probabilities $\widehat{\xi}_{t|\tau}$ when $\tau = t$,
  - *smoothed* probabilities $\widehat{\xi}_{t|\tau}$ when $t < \tau \leq T$.